



REALITY CHECK ASSESSMENT BOARD

General Certificate of Advanced Level

PURE MATHEMATICS

6042/1

PAPER 1

2 hours

Revision Test 7 August 2020

Additional Materials :

Answer paper

List of Formulae MF7

Scientific calculator

Graph paper

TIME 2 hours 20 minutes

INSTRUCTION TO CANDIDATES

Write your name in the spaces provided on the answer sheet/answer booklet.

Answer all questions

If a numerical answer cannot be given exactly, and the accuracy required is not specified in the question, then in the case of an angle it should be given to the nearest degree, and in other cases it should be given correct to 2 significant figures.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question. The total number of marks for this paper is 90.

Questions are printed in the order of their mark allocations and candidates are advised to attempt questions sequentially.

The use of an electronic calculator is expected, where possible.

You are reminded of the need for clear presentations in your answers

This paper contains 3 printed pages

Created by Mabauwa 0719 275 379

Copyright RCAB2020

1 Solve the equation $\log_x 4 + \log_2 x = 3$. [5]

2
$$f(x) = \frac{16}{(1-x)(2-x)^2} .$$

(a) Express $f(x)$ in partial fractions . [4]

(b) Hence show that if x is sufficiently small then $f(x) \approx 4 + 8x + 11x^2$. [4]

3 Solve the inequality $3 - |x - 2| < |4 - 2x| + 2$. [6]

4 The lines l_1 and l_2 have equations $x + 4y + 5 = 0$ and $y = 2x - 2$ respectively .

(a) Find the coordinates of A the point of intersection l_1 and l_2 . [2]

(b) Find the equation of a line through A which is perpendicular to l_1 . [3]

5 When the polynomial $4x^3 - 2x^2 + mx + n$ is divided by $x^2 + 2x - 5$ there is no remainder .

Find the value of m and the value of n . [5]

6

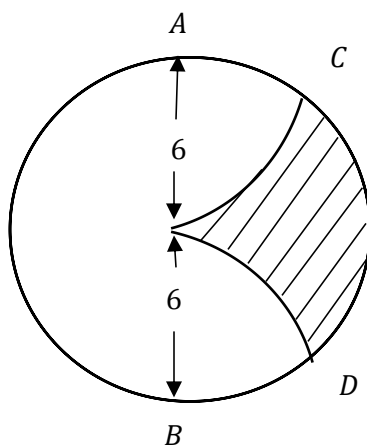


figure 1

Figure 1 shows a circle centre O of radius 6 cm . An arc OC with centre at A is drawn with radius 6 cm . A second arc OD with centre at B is drawn with radius 6 cm .

Show that the shaded region OCD is $6(3\sqrt{3} - \pi)$. [5]

7 (a) Oil leaking from car is forming a circular oil spillage on the ground , whose area is increasing at a steady rate of $360m^2s^{-1}$.

Find the rate at which the radius is increasing a minute after is started forming . [4]

(b) A closed rectangular box has a square base of x cm and height of y cm . Its total surface area is $400cm^2$.

(i) Find an expression of y in terms of x . [3]

(ii) Find the largest possible volume of the box . [4]

- 8 Find the equation of a circle which touches the y axis at $y = 3$ and passes through the point $5, 2$. [6]
- 9 The points O, A, B and C form a parallelogram where O is the origin. The position of A and C are $\mathbf{i} + \lambda\mathbf{j} + \mathbf{k}$ and $4\mathbf{i} + 2\mathbf{k}$ respectively, where λ is positive constant. The angle $\widehat{OAC}$ is a right angle.
- Find (a) the value of λ . [2]
- (b) the position vector of B . [3]
- (c) the exact area of the parallelogram $OABC$. [4]
- 10 (a) Prove the identity $\tan(x + 60^\circ) \tan(x - 60^\circ) = \frac{\tan^2 x - 3}{1 - 3 \tan^2 x}$. [4]
- (b) Solve the equation $\tan(\theta + 45^\circ) = 1 - 2 \tan \theta$, $0 \leq \theta \leq 360$. [6]
- 11 Given that a geometric progression has first term 10 and common ratio r greater than 0. The sum of the first seven terms is three times greater than the 8^{th} and 9^{th} terms.
- Show that $r^7(4 - 3r^2) = 1$. [5]
- 12 By using the substitution $x = \operatorname{cosec} \theta$, show that
- $$\int_{\sqrt{2}}^2 \frac{\sqrt{x^2 - 1}}{x} dx = \sqrt{3} - 1 - \frac{\pi}{12}. \quad [6]$$
- 13 A cylindrical tank has constant radius of 0.9 m . The volume, $V \text{ m}^3$, of the water in the tank has height $h \text{ m}$. Water can be poured into the tank from a tap at the top of the tank and can be drained out of a tap at the base of the tank, which are initially both turned off. Water then starts pouring in at the constant rate of $0.36\pi \text{ m}^3$ per minute and at the same time water begins to drain out at the rate of $0.45\pi h \text{ m}^3$ per minute.
- (a) Given further that t is measured when both taps are turned on, show that
- $$9 \frac{dh}{dt} = 4 - 5h. \quad [4]$$
- Initially the water in the tank has height of 4 m .
- (b) Solve the differential equation above and show that $h = \frac{4}{5} \left(1 + 4e^{-\frac{5}{9}t} \right)$. [5]

High voltage